

ZITIAntao Lin

Boston, MA — lin.ziti@northeastern.edu — linkedin.com/in/zitiantao-lin

EDUCATION

Northeastern University

Master of Science in Information Systems

Boston, MA

Jan 2025 – Dec 2026 (Expected)

Graduate Research Assistant at CRAFT Lab (Cognitive Robotics & AI for Future Technology)

Teaching Assistant for INFO 6105: Data Science Engineering Methods

Focus: Machine Learning Systems, Data Pipelines, Embodied AI, Human-Robot Interaction

Civil Aviation Flight University of China

Bachelor of Engineering in Information and Computing Science

Sichuan, China

Sep 2020 – Jul 2024

Strong foundation in programming, algorithms, numerical computation, and machine learning

Research projects in data science, time-series modeling, and computer vision

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, C#, MATLAB, LaTeX

Machine Learning: Large Language Models (LLMs), Multimodal Learning, Computer Vision, Transformer Models, Time Series Forecasting, Depth Estimation

Frameworks & Tools: PyTorch, ROS, Unity, XR / Mixed Reality, Git, Docker, TensorRT

Systems: Linux, HPC (Slurm / Explorer), Data Pipelines, Modular ML Systems

PROFESSIONAL EXPERIENCE

Northeastern University — CRAFT Lab

Graduate Research Assistant (Advisor: Prof. Gilbert Yang Ye)

Boston, MA

Feb 2025 – Present

Design and implement end-to-end embodied AI systems integrating perception, learning, and robotic control

Develop multimodal machine learning pipelines combining vision, gaze signals, and robotic manipulation

Build XR / Unity simulation environments for evaluating human-robot interaction systems

Integrate large language model components for high-level reasoning in robotic interaction tasks

Conduct experimental evaluation and data-driven analysis to validate system performance

Northeastern University

Teaching Assistant — INFO 6105: Data Science Engineering Methods

Boston, MA

Jan 2026 – Present

Support a graduate-level course covering data engineering pipelines and machine learning systems

Assist students with Python programming, feature engineering, and model implementation

Review and debug student code while providing structured feedback on algorithm efficiency

Lead office hours to support debugging and ML workflow design for over 100 students

Beijing Sohu New Media Information Technology Co Ltd

Software Engineering & Product Operations Intern

Beijing, China

Mar 2021 – Aug 2021

Contributed to development and optimization of a large-scale mobile application platform

Conducted funnel analysis and user behavior analysis to identify usage bottlenecks

Designed data-driven strategies for feature iteration and user retention optimization

Maintained modular software components supporting scalable development pipelines

RESEARCH PROJECTS

GGC-Trans: Gaze-Guided Cross-View Transformer for Calibration-Free Depth Generation

Dec 2025 – Present

Proposed a Transformer architecture bridging egocentric and exocentric visual representations

Introduced gaze heatmaps as attention priors for cross-view spatial reasoning

Designed hybrid loss combining L1 regression and gradient-based edge consistency

Built a depth supervision pipeline using RealSense RGB-D ground truth

BEAR: Benchmarking and Enhancing Multimodal LLMs for Atomic Embodied Capabilities

Jun 2025 – Sep 2025

Contributed to a large-scale benchmark evaluating embodied multimodal language models

Implemented baseline models and conducted diagnostic evaluation of perception and reasoning

Co-developed BEAR-Agent integrating visual and spatial reasoning tools

GazeCastGrasp: MR-Based Eye-Gaze Interaction for Assistive Robotic Manipulation

Jul 2025 – Sep 2025

Developed an end-to-end gaze-driven robotic manipulation system in mixed reality

Integrated eye-tracking, object perception, and ROS robotic control

Conducted user studies comparing gaze-based interaction with traditional input methods

Eye-Gaze Interaction System for Wearable MR-Based Robotic Manipulation

Feb 2025 – Jul 2025

Built a first-person gaze interaction pipeline integrating MR visualization and robot control

Implemented real-time object recognition for gaze-based target localization

Enabled hands-free robotic object manipulation using gaze fixation

LSTM-Based Time Series Forecasting for Civil Aviation Passenger Traffic

Jan 2023 – Dec 2023

Collected and processed aviation passenger traffic data from 2006–2022

Developed LSTM models capturing long-term temporal patterns

Compared deep learning forecasts with traditional statistical baselines

PUBLICATIONS

RaycastGrasp: Eye-Gaze Interaction with Wearable Devices for Robotic Manipulation

IEEE International Conference on Intelligent Robotics (ICIR 2025)

Best Paper Award

BEAR: Benchmarking and Enhancing Multimodal Language Models for Atomic Embodied Capabilities

Under Review, 2026

RESEARCH INTERESTS

Embodied AI, Multimodal Machine Learning, Human–Robot Interaction, Computer Vision, Eye-Gaze Interaction, Robotic Manipulation, Cross-View Perception, Intelligent Interactive Systems